

Data Profiler – Theory Concepts

1. What is Data Analysis?

Data Analysis is the systematic process of inspecting, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision-making. It combines statistics, domain knowledge, and computing to convert raw, messy data into actionable insight.

Core stages of data analysis:

- Collection — gathering data from files, databases, APIs, or sensors.
- Cleaning — handling missing values, duplicates, and inconsistent formats.
- Exploration — understanding distributions, relationships, and anomalies (EDA).
- Modeling / Interpretation — applying statistical or ML techniques.
- Communication — presenting findings via visualizations, reports, dashboards.

2. How to Plan a Data Science Project

A data science project follows a structured lifecycle, commonly summarized as CRISP-DM (Cross-Industry Standard Process for Data Mining), adapted below into practical steps:

1. Business Understanding — Define the problem in business terms and set success metrics.
2. Data Understanding — Identify available data, its sources, and quality.
3. Data Acquisition — Collect and consolidate data from all relevant sources.
4. Data Preparation / Cleaning — Handle missing values, duplicates, incorrect types, irrelevant fields.
5. Exploratory Data Analysis — Use univariate, bivariate, multivariate analysis to find patterns.
6. Feature Engineering — Create, transform, or select variables to improve model performance.
7. Modeling — Frame and train an appropriate ML model.
8. Evaluation — Validate the model against business goals with proper metrics.
9. Deployment & Monitoring — Ship the model to production and monitor performance/drift.
10. Documentation & Communication — Document the process and communicate insights to stakeholders.

3. Machine Learning Problem Statement

Problem Type: Binary Classification

Goal: Predict whether a customer will churn (churn = 1) or stay (churn = 0), based on their demographic profile, purchase behavior, membership history, and engagement metrics.

Target Variable: churn (0 = retained, 1 = churned)

Business Value: Identifying churn-risk customers early lets the business trigger targeted retention campaigns, reducing revenue loss.

4. What are Tensors?

A tensor is a generalized, multi-dimensional array of numbers — the core data structure used to represent inputs, outputs, and parameters in machine learning and deep learning frameworks (NumPy, PyTorch, TensorFlow). Tensors are classified by their **rank** (number of dimensions/axes).

Rank	Name	Example	Shape
0	Scalar	A single number, e.g. 42	()
1	Vector	One customer's feature list	(n,)
2	Matrix	Full dataset (rows x columns)	(rows, cols)
3+	N-D Tensor	A batch of matrices (e.g. batch of customer records)	(batch, rows, cols)

NumPy example — tensors of increasing rank:

```
import numpy as np
```

```
scalar = np.array(42) # rank-0, shape () vector = np.array([34, 55000, 6, 4200.0]) # rank-1, shape (4,) matrix = np.array([[34, 55000, 6, 4200.0], [29, 48000, 3, 1500.0]]) # rank-2, shape (2, 4) tensor_3d = np.stack([matrix, matrix * 1.05]) # rank-3, shape (2, 2, 4)
```

Why tensors matter: the cleaned customer table itself is a rank-2 tensor (rows = customers, columns = features). Deep learning models batch such tensors into rank-3+ structures for efficient, parallel GPU computation, and every core ML operation (dot products, matrix multiplication, gradients) is defined on tensors.

5. Data Profiling

Data profiling is the automated process of examining a dataset to summarize its structure, quality, and content — producing statistics on missing values, data types, distributions, correlations, and potential quality issues (duplicates, high cardinality, skew) before it is used for modeling. In this project, an automated profiling report was generated (see [reports/pandas_profiling_report.html](#)) covering:

- Missing value counts and percentages per column.
- Descriptive statistics (mean, median, std, min/max, quartiles).
- Correlation matrices between numerical variables.
- Automated warnings on skew, high cardinality, duplicate rows, and constant columns.

6. Data Cleaning Techniques Applied

- **Missing data:** numerical columns (age, income) imputed with the column median, which is robust to outliers.
- **Inconsistent types:** categorical columns (gender, region, membership_tier) cast to the pandas 'category' dtype; count-like columns cast to int.
- **Duplicates:** exact duplicate customer_id rows introduced by the raw CSV export were identified and removed.
- **Irrelevant columns:** reviewed for redundancy against the churn ML problem before finalizing the feature set.

7. Exploratory Data Analysis (EDA) Types

- **Univariate Analysis** — examines the distribution of a single variable (e.g., histogram of Age, Income, Total Spend).
- **Bivariate Analysis** — examines the relationship between two variables (e.g., Gender vs. Purchases, Income vs. Churn).
- **Multivariate Analysis** — examines relationships among three or more variables simultaneously (e.g., correlation heatmaps, pair plots).